8.

factors = types of fertilizers levels: type A, type B, type C

treatment: type A, type B, type C

d. factors = daily dose of vitamin C levels: 200 mg, 500 mg, 1000 mg

treatment: daily dose vitamin C = 200 mg, daily dose vitamin C = 500 mg, daily dose vitamin C = 1000 mg

14.

assume normal:
① $n = 25$, large enough to use C.L.T → normal distribution
② proportion is binominal distribution, when n large → bell-shape

assume not normal: If proportion close to 0 or 1, may not normal distribution

8. $n = 11$, $k = 3$, $\bar{x}_1 = 76$, $\bar{x}_2 = 66.33$, $\bar{x}_3 = 87$

a) $CM = \frac{(840)^2}{11} = \frac{705600}{11} = 64145.45$

$TSS = 65286 - 64145.45 = 1140.55$

$SST = (5 \times 76^2) + 3 \times (66.3)^2 + 3(87)^2 - 64145.45 = 641.87$

$SSE = 1140.55 - 641.87 = 511.93$

| Source | SS | df | MS | F |
|---|---|---|---|---|
| Treatment | 641.87 | 3-1=2 | MST = 320.94 | 5.1487 |
| Error | 498.67 | 11-3=8 | MSE = 62.33375 | |
| Total | 1140.55 | 11-1=10 | | |

$H_0: \mu_1 = \mu_2 = \mu_3$

$H_a: \mu_i \neq \mu_j$, if $i \neq j$

$\alpha = 0.05$

$F^* = 5.1487$

$F_{2,8,0.05} = 4.46$

$\therefore F^* > F_{2,8,0.05}$, so reject $H_0$.

则能明白的证据说各个M不一样。

10. a) $H_0: \mu_1 = \mu_2 = \mu_3$

$H_a: \mu_i \neq \mu_j, i \neq j$

$F^* = 4.256$, $\alpha = 0.01$

p-value $= P(F_{2,9} > 4.256) =$ between $0.01 \sim 0.05$

$\therefore$ so p-value $> \alpha$, non-reject $H_0$

b) $(\bar{x}_1 - \bar{x}_2) \pm t_{9, 0.005} \sqrt{\frac{MSE}{n_1} + \frac{MSE}{n_2}}$

$= (60.5 - 54.667) \pm 3.25 \sqrt{(14.941)(\frac{1}{4} + \frac{1}{3})}$

$= 5.833 \pm 3.25 (2.95221)$

$= [-3.7017, 15.4277]$

c) $\bar{x}_1 \pm t_{9, 0.005} \left( \sqrt{\frac{14.941}{4}} \right)$

$= 60.5 \pm 3.25 \left( \sqrt{\frac{14.941}{4}} \right)$

$= [54.2188, 66.7812]$

d) i) no outlier in each group

ii) shape not skew & is symmetric, probably is normal population.

14) $n = 37, k = 4$

a) $CM = \frac{1624.9961}{37} = 43.9161$

$TSS = 51.6489 - 43.9161 = 7.7328$

$SST = 44.9385 - 43.9161 = 0.9224$

$SSE = 7.7328 - 0.9224 = 6.8104$

| Source | SS | df | MS | F |
|---|---|---|---|---|
| Treatment | 0.9224 | 3 | 0.3075 | 1.4898 |
| Error | 6.8104 | 33 | 0.2064 | |
| Total | 7.7328 | 36 | | |

b) $H_0: \mu_1 = \mu_2 = \mu_3 = \mu_4$

$H_a: \mu_i \neq \mu_j, i \neq j$

$\alpha = 0.05$

$F_{STAT} = \frac{MST}{MSE} \sim df_1, df_2$

$F^* = 1.4898$

RR: $\alpha = 0.05$

$F_{3,33,0.05} = 2.9016 > 1.4898$

$\therefore$ non-reject $H_0$

$\alpha = 0.01$

$F_{3,33,0.01} = 4.4769 > 1.4898$

$\therefore$ non reject $H_0$.

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(Contd) 14)

c) $(0.8964 - 1.1472) \pm t_{33,\,0.025}\sqrt{0.2064\left(\frac{1}{14}+\frac{1}{11}\right)}$

$= (0.8964 - 1.1472) \pm 2.035\sqrt{0.2064\left(\frac{1}{14}+\frac{1}{11}\right)}$

$= [-0.6214,\ 0.1216]$

d) $(1.28 - 1.225) \pm 2.035\sqrt{0.2064\left(\frac{1}{8}+\frac{1}{9}\right)}$

$= [-0.5112,\ 0.6212]$

e) Light tuna in water, white tuna in water

Light tuna in water, white tuna in oil

white tuna in water, light tuna in oil

white tuna in oil, light tuna in oil

$C^4_2 = \frac{4\times 3}{2\times 1} = 6$

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